

WELL COMPLETION REPORT

Synthetic Field Cluster — SYN-Assam-Duliajan (Upper-Assam-type setting)

1. WELL IDENTIFICATION

Well ID	W006
Well Name	SYN-Assam-Duliajan-006
Field / Cluster	SYN-Assam-Duliajan
Area	Synthetic Assam Field Cluster (Duliajan-type operating area)
Latitude	27.342534 N
Longitude	95.381181 E
Well Status	Abandoned
Well Role	Historical (offset reference well)
Spud Date	11-Sep-2019
Drilling Completed	24-Sep-2019
Report / Completion Date	28-Sep-2019
Total Depth (Driller)	3,960 m
Primary Formation	Sandstone
Formations Penetrated	Claystone / Shale / Sandstone / Limestone
Rig	SYN-RIG-E1400-04
Drilling Days	14 days
Recorded Events	0
Recorded NPT	0.0 hours

2. MACHINE-READABLE HEADER BLOCK

Document Type: Well Completion Report
Well ID: W006
Well Name: SYN-Assam-Duliajan-006
Field: SYN-Assam-Duliajan
Latitude: 27.342534
Longitude: 95.381181
Total Depth: 3,960 m
Primary Formation: Sandstone
Well Status: Abandoned
Spud Date: 2019-09-11
Event Count: 0
Total NPT: 0.0 hours

PROTOTYPE TRAINING RECORD. Synthetic demonstration document prepared for the eRTMAC-NWIS prototype (SIH PS 26121). The well, coordinates, formations, events and measurements described here are generated for software demonstration and are not an operational record of any company. Location names are used for interface readability only.

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3. WELL SUMMARY

3.1 Objective

Well W006 (SYN-Assam-Duliajan-006) was drilled to provide an additional development penetration of the primary sandstone objective and confirm the area formation sequence. The well was drilled vertically from surface to a driller's total depth of 3,960 m within the SYN-Assam-Duliajan synthetic cluster.

3.2 Drilling Interval

Surface to 3,960 m, completed in 14 rotary drilling days between 11-Sep-2019 and 24-Sep-2019. Average overall penetration rate for the well was 11.8 m/hr on a 24-hour basis.

3.3 Formations Encountered

The well penetrated 4 formation units: Claystone (0–821 m), Shale (821–2,260 m), Sandstone (2,260–3,521 m), Limestone (3,521–3,960 m). The primary objective formation was Sandstone.

3.4 Major Operational Events

No reportable drilling events were recorded for this well. All hole sections were drilled, cased and cemented without significant non-productive time. This well is retained in the offset database as a **low-risk reference penetration**.

3.5 Casing Operations

4 casing strings were run: 13-3/8" at 820 m, 9-5/8" at 2,260 m, 7" at 3,720 m, 5" liner at 3,960 m. All strings were cemented as detailed in Section 7.

3.6 Completion Status

Well status at report date: **Abandoned**. The well was plugged and abandoned after evaluation; cement plugs were set across the open hole and shoe intervals.

4. OFFSET-WELL CONTEXT

The following synthetic wells of the same cluster lie within short offset distance and share a comparable formation sequence. They are the primary comparison set for offset-well analysis.

Well ID	Well Name	Distance (km)	TD (m)	Primary Formation	Events
W003	SYN-Assam-Duliajan-003	5.24	3,850	Sandstone	8
W001	SYN-Assam-Duliajan-001	5.84	3,470	Sandstone	4
W009	SYN-Assam-Duliajan-009	6.18	3,180	Sandstone	6
W004	SYN-Assam-Duliajan-004	8.40	3,590	Sandstone	2
W008	SYN-Assam-Duliajan-008	10.10	3,690	Sandstone	8
W002	SYN-Assam-Duliajan-002	10.52	3,450	Sandstone	7

5. FORMATION / DEPTH RECORD

Formation tops and bases as recorded from cuttings and log correlation. Depths are driller's depths below rotary table.

Unit	Formation	Top Depth (m)	Bottom Depth (m)	Thickness (m)	Remarks
1	Claystone	0	821	821	Routine drilling, no reportable event
2	Shale	821	2,260	1,439	Routine drilling, no reportable event
3	Sandstone	2,260	3,521	1,261	Routine drilling, no reportable event
4	Limestone	3,521	3,960	439	Routine drilling, no reportable event

Formation: Claystone Top: 0 m Bottom: 821 m
Formation: Shale Top: 821 m Bottom: 2260 m
Formation: Sandstone Top: 2260 m Bottom: 3521 m
Formation: Limestone Top: 3521 m Bottom: 3960 m

6. DRILLING PARAMETERS BY FORMATION

Representative values summarised from the depth-indexed parameter record for this well. Mud weight is reported in ppg, rate of penetration in m/hr, rotary speed in rpm and weight on bit in klbf.

Formation	Interval (m)	Mud Wt (ppg)	MW range	ROP (m/hr)	RPM	WOB (klbf)
Claystone	0–821	9.58	9.0–10.2	31.8	141	8.1
Shale	821–2,260	11.28	10.2–12.3	15.8	123	18.8
Sandstone	2,260–3,521	13.00	12.1–13.9	14.7	107	27.7
Limestone	3,521–3,960	14.56	14.3–14.9	7.0	85	38.3

6.1 Mud Programme Summary

Water-based inhibited mud was used throughout. Mud weight was raised progressively with depth from approximately 9.6 ppg in the upper section to 14.6 ppg at total depth. No loss-related mud weight reductions were required.

6.2 Bit / Hydraulics Note

Rotary speed was held in the 118 rpm range on average with weight on bit averaging 21.7 klbf. Penetration rate declined with depth from 31.8 m/hr in the upper hole to 7.0 m/hr at total depth, consistent with the area trend.

7. DRILLING EVENTS RECORD

No reportable drilling events were recorded for well W006. All hole sections were drilled within the planned parameter envelope.

Well ID: W006
Event Count: 0
Event Type: None recorded
Total NPT: 0.0 hours
Risk Classification (offset reference): LOW

The absence of recorded events across the 3,960 m of section drilled makes this well a useful negative control within the offset database: it demonstrates that the formation sequence of this cluster can be drilled without incident when the parameter envelope described in Section 6 is maintained.

Formation	Interval (m)	Recorded Events	Remarks
Claystone	0–821	None	Routine drilling
Shale	821–2,260	None	Routine drilling
Sandstone	2,260–3,521	None	Routine drilling
Limestone	3,521–3,960	None	Routine drilling

8. CASING AND CEMENTING RECORD

Casing	Grade / Weight	Shoe Depth (m)	Hole Size	Cement Programme
13-3/8"	K-55, 68 lb/ft	820	17-1/2"	Lead 12.6 ppg / Tail 15.8 ppg, cement to surface
9-5/8"	N-80, 47 lb/ft	2,260	12-1/4"	Lead 12.9 ppg / Tail 15.8 ppg, TOC 700 m
7"	N-80, 29 lb/ft	3,720	8-1/2"	Tail 15.8 ppg, TOC 2,110 m
5" liner	N-80, 18 lb/ft	3,960	6-1/8"	Tail 15.8 ppg, liner lap tested

8.1 Cementing Observations

All casing strings were run and landed without restriction. Cement returns and bond logs were within acceptance criteria for every stage; no remedial cementing was required.

8.2 Pressure Testing

Each string was pressure tested after landing and formation integrity tests were carried out below the 13-3/8" and 9-5/8" shoes. Recorded results were satisfactory except where noted in Section 7.

9. FINAL SUMMARY

Final Depth	3,960 m
Primary Formation	Sandstone
Major Events	None recorded
Total NPT	0.0 hours
Completion Status	Abandoned
Offset Risk Classification	LOW — no recorded events

9.1 Drilling Challenges

No significant drilling challenges were encountered.

9.2 Key Historical Observations

- The full formation sequence of this cluster was drilled without incident, confirming that the recorded parameter envelope is workable for this area.
- Formation tops in this well correlate closely with the neighbouring wells listed in Section 4, supporting depth-based offset comparison across the cluster.

9.3 Lessons Learned

- This well can be used as the baseline parameter reference for the cluster; deviations from its recorded envelope in future wells should be treated as an early indicator.
- No remedial operations were required, so the casing and cement programme in Section 8 is recommended as the area default.

10. DOCUMENT CONTROL

Document Number	WCR-W006-R0
Revision	R0
Issue Date	28-Sep-2019
Prepared By	Drilling Engineering (prototype dataset generator)
Classification	Prototype Training Record — synthetic demonstration data
Source Datasets	wells.csv, well_formations.csv, drilling_events.csv, drilling_parameters.csv

All identifiers, coordinates, depths, formations, events and measurements in this document are synthetic and were generated for the eRTMAC-NWIS prototype. This document is not an operational record of Oil India Limited or any other operator, and no confidential information is contained in it.